Workshop on Advanced Sampling Methodologies

Introduction to R Programming

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Workshop's agenda

- **Day 1**: Introduction to QGIS
- **Day 2**: Automatic update of census EA and national sampling frame
- **Day 3**: Introduction to R programming
- **Day 4**: Introduction to Sampling Design
- **Day 5**: Advanced R programming
- **Day 6**: Earth observation (EO) for sampling design
- **Day 7**: AI and machine learning for sample selection
- **Day 8**: Simulation and optimization of sampling strategies
- **Day 9**: R Shiny for sampling design evaluation
- **Day 10**: Wrap-up and workshop evaluation

Today's agenda

- Introduction to R and RStudio
- Data types and data structures
- Working with scripts and the console
- Importing data
- Saving and closing sessions
- Practical exercises

What is R?

Why R?

- Powerful tool for **data analysis and visualization**
- Encourages **reproducibility** and **sharing**
- Thousands of user-created **packages** extend its functionality
- Large and active **community** with extensive support

Iris k-means clustering

X Variable

Sepal.Length

Y Variable

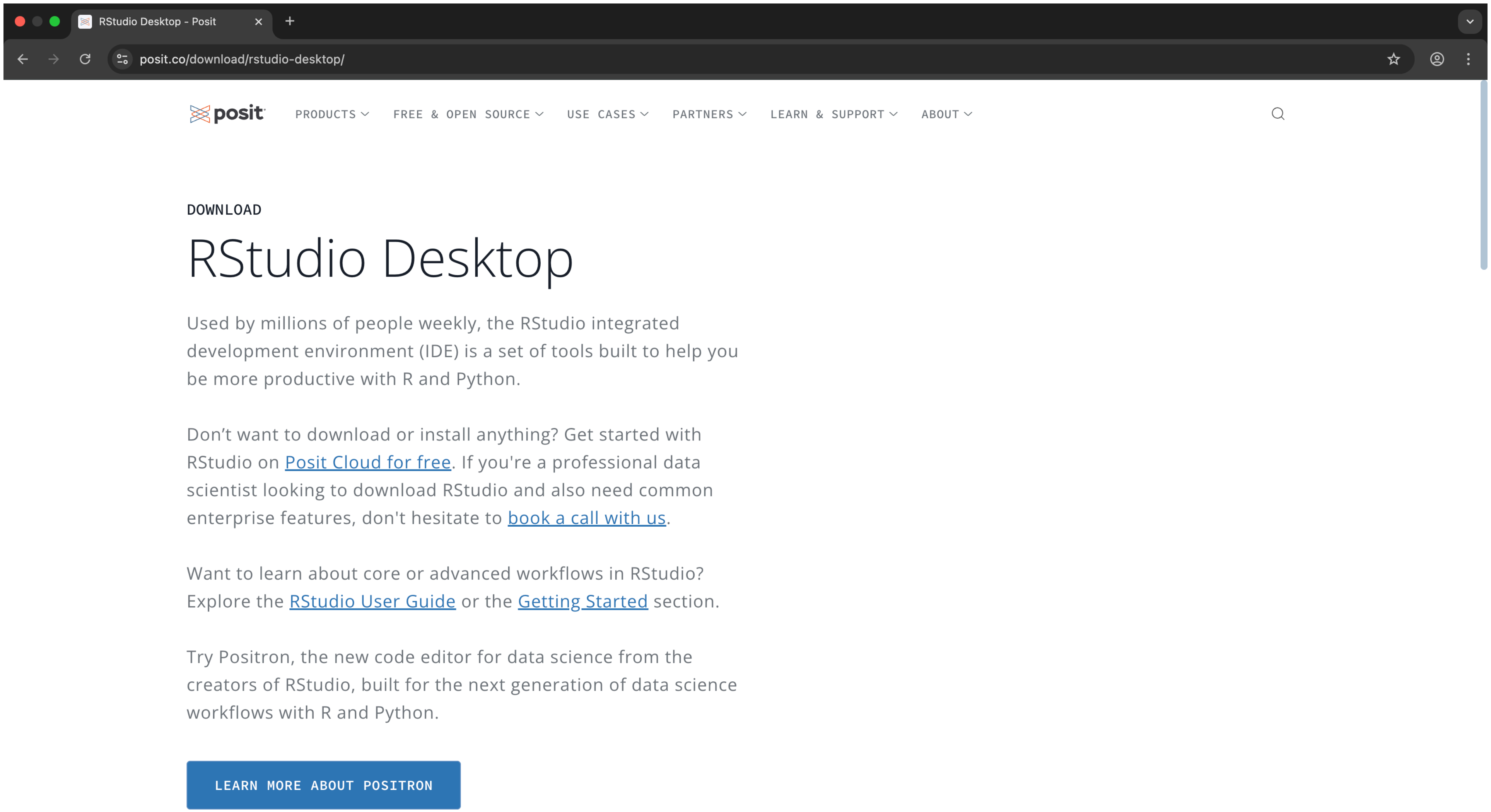
Sepal.Width

Cluster count

3

Installing R and RStudio

- Download and install **R and RStudio** from posit.co


RStudio Desktop - Posit

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RStudio Desktop

Used by millions of people weekly, the RStudio integrated development environment (IDE) is a set of tools built to help you be more productive with R and Python.

Don’t want to download or install anything? Get started with RStudio on Posit Cloud for free. If you're a professional data scientist looking to download RStudio and also need common enterprise features, don't hesitate to book a call with us.

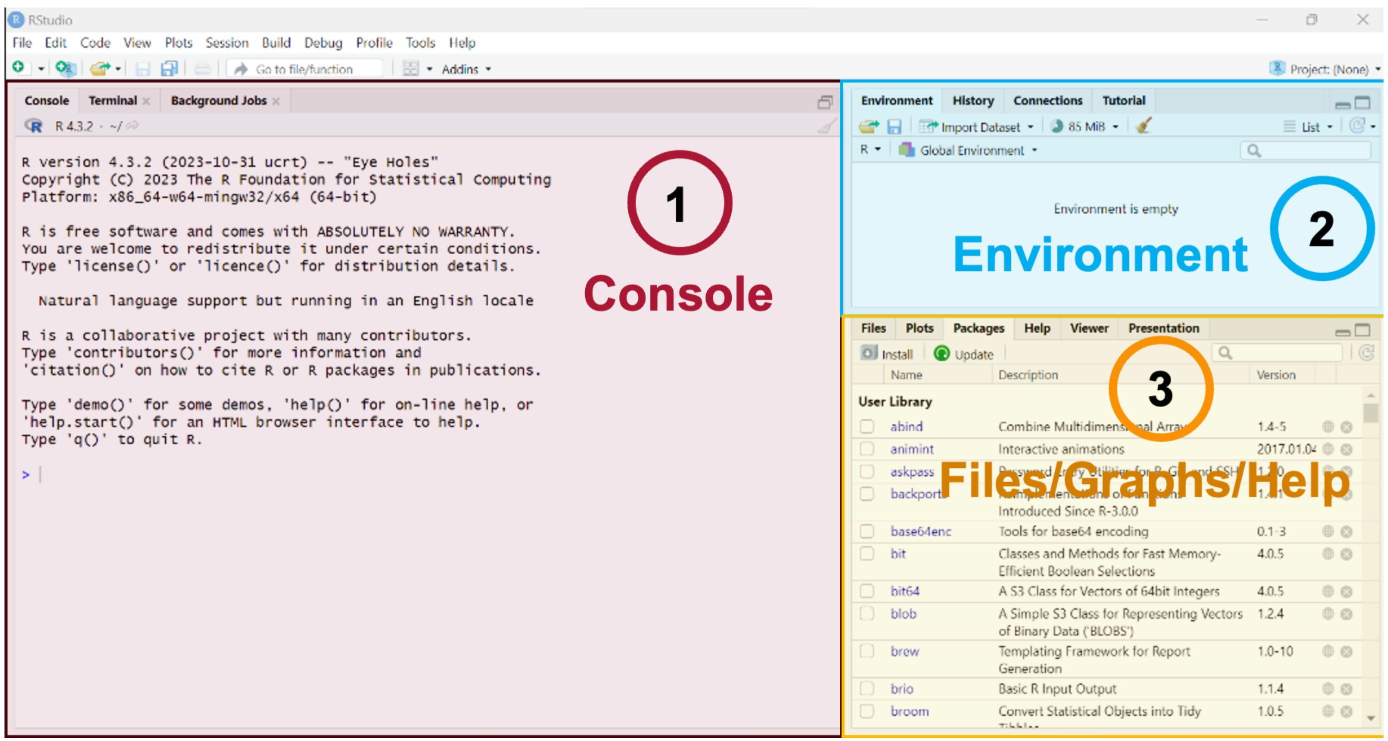
Want to learn about core or advanced workflows in RStudio? Explore the RStudio User Guide or the Getting Started section.

Try Positron, the new code editor for data science from the creators of RStudio, built for the next generation of data science workflows with R and Python.

LEARN MORE ABOUT POSITRON

RStudio interface

RStudio has four main panes:
- **Source Editor** – write and save code
- **Console** – run commands interactively
- **Environment/History** – see and manage objects
- **Files/Plots/Packages/Help/Viewer** – tools and outputs



Script Editor vs Console

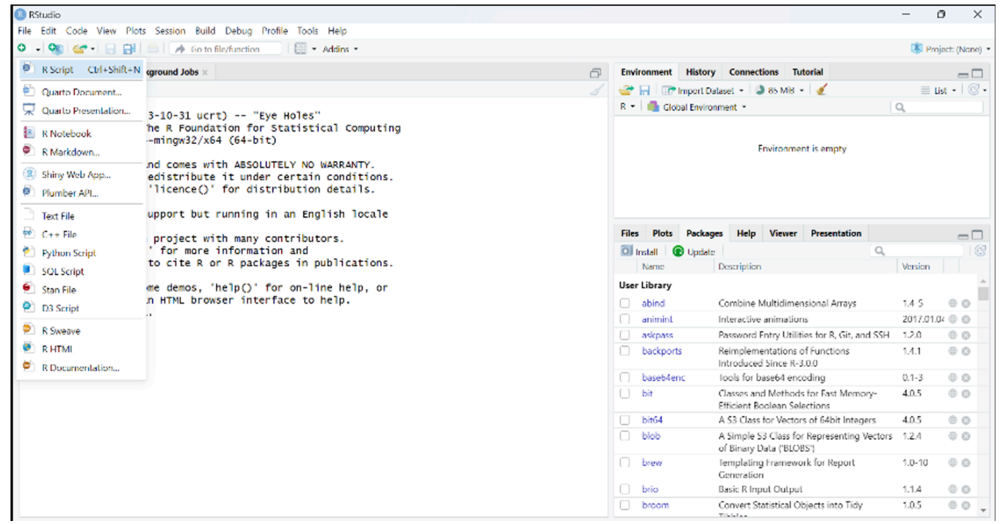
Script Editor

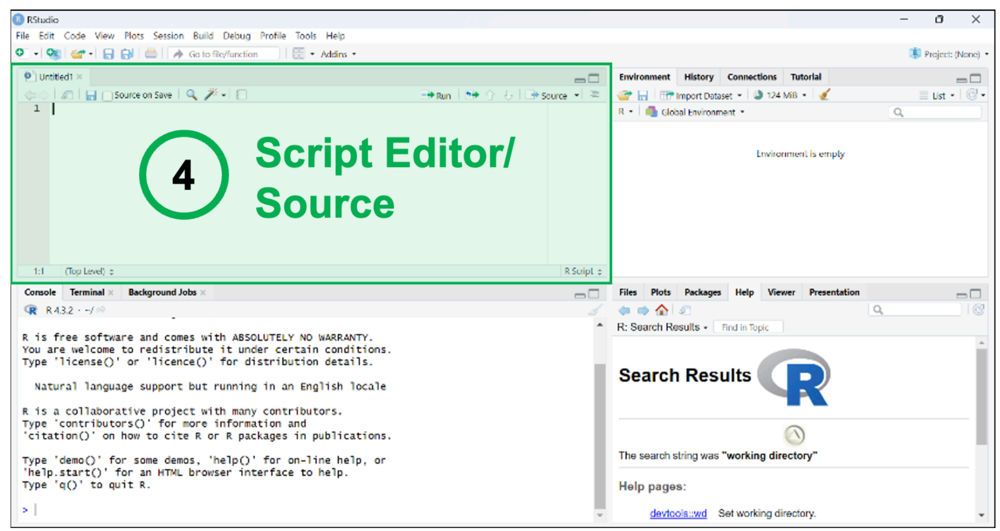
- Write, edit, and save code
- Organize analysis in reproducible scripts
- Prevent accidental execution before editing

Console

- Executes code immediately
- Use for quick checks and interactive testing
- Output and errors appear here

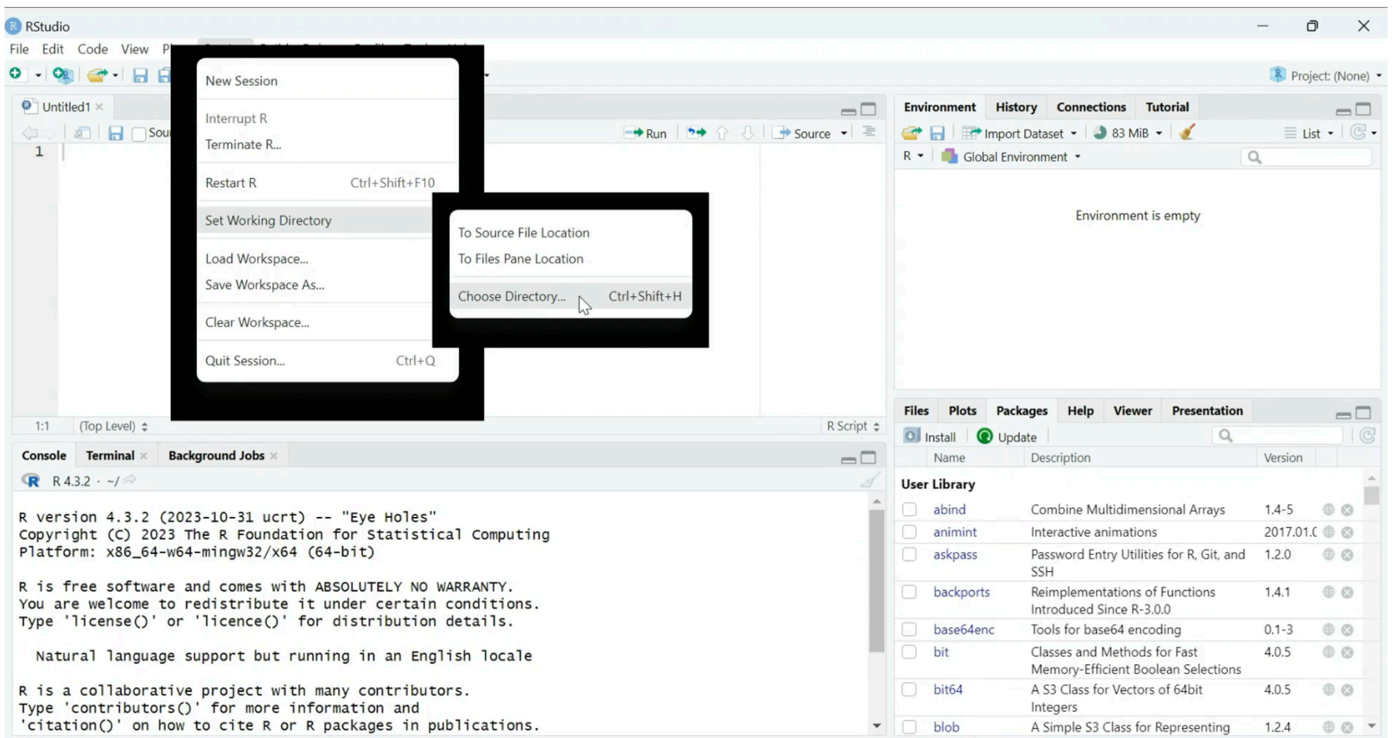
ℹ️ Best practice: write in the Script Editor, run via Console.





Working Directory

- Your working directory is the folder where R looks for and saves files.
- Set with: `setwd("path")`
- Check with: `getwd()`
- Good practice: create a dedicated folder for each project.



ℹ️ Try to run the code in your R Console and in the Script Editor.

R as a calculator

- R can perform arithmetic directly

```r
1 + 2 * 8
a <- log(10)
b <- 5
b + b
```

- Results in Console are prefixed with `[1]` (index of first element).
- `[2], [3], ...` appear if output spans multiple lines.

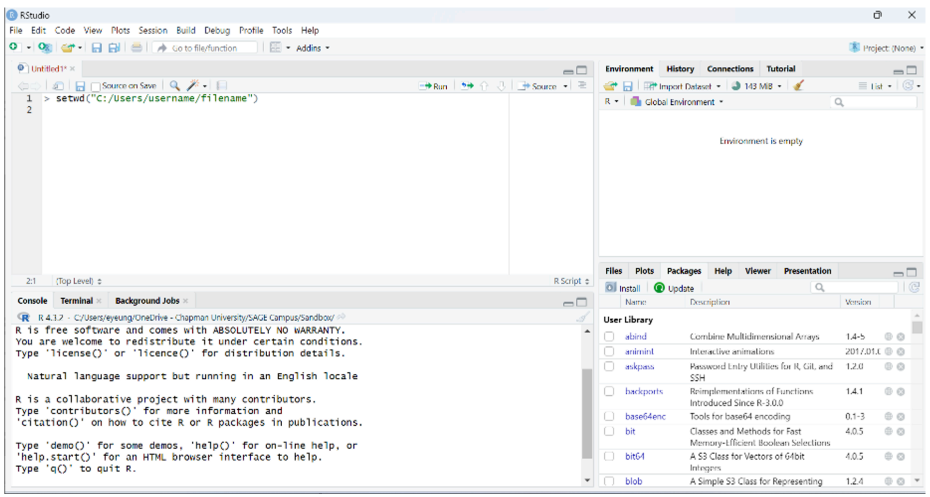
ℹ️ Try to run the code in your R Console and in the Script Editor.

R packages

- Collections of functions, datasets, and documentation.
- Expand R beyond base functions.
- Install from CRAN, GitHub, or other repositories.

In RStudio

- Use the *Packages* tab to install, update, or remove.

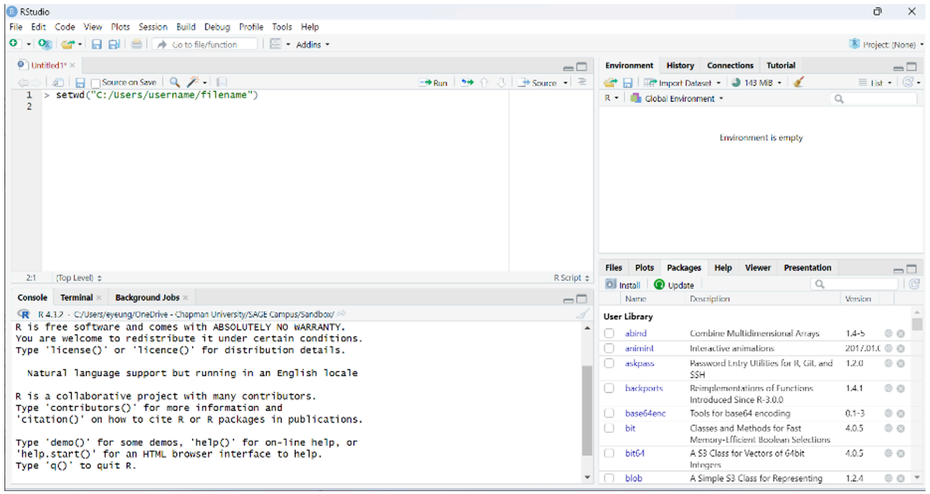


- Or install in the Script Editor

```r
install.packages("dplyr")
```

Loading packages

- Load packages into a session via checkbox in the *Packages* pane.



- Or load packages into a session with:

```r
library(dplyr)
```

ℹ️ Best practice: only load what you need.

R objects

- Everything in R is an **object** (vectors, lists, functions, data frames, etc.).
- Assign values using `<-`.

```
x <- 5
x + x    # returns 10
```

Naming tips
- ✅ Descriptive and concise
- ✅ Use underscores `_` or periods `.`
- ❌ Avoid very long names
- ❌ Avoid existing function names
- ❌ No blank spaces

ℹ️ Try to run the code in your Script Editor.

Data types

- **Numeric**: numbers, may include decimals.

```
w <- c(1.7, 3)
x <- c(1, 2, 3)   # integers are numeric without decimals
```

- **Character**: text strings.

```
y <- c("male", "female")
```

- **Logical**: TRUE/FALSE values.

```
y <- c(TRUE, FALSE)
```

- **Factor**: categorical variables.

```
gender <- factor(c("boy", "girl", "girl", "boy"))
```

Check type:

```
class(object)
typeof(object)
```

Data structures

- **Vectors**: one-dimensional, homogeneous elements
- **Matrices**: two-dimensional, homogeneous elements
- **Lists**: flexible, heterogeneous elements
- **Arrays**: multi-dimensional, homogeneous elements
- **Data frames**: tabular, columns may differ in type

Vectors

- The most basic structure in R
- One-dimensional, elements must be the **same type**

```r
numbers <- c(1, 2, 3, 4, 5)
colors <- c("red", "green", "blue")
logical_values <- c(TRUE, FALSE, TRUE)
```

Indexing with brackets:

```r
numbers[1]       # first element
colors[2:3]      # second and third elements
```

Vectorized operations apply to all elements:

```r
numbers * 2      # multiplies each element by 2
```

ℹ️ Try to run the code in your Script Editor.

Matrices

- Two-dimensional, **homogeneous elements**
- Create with `matrix()`

```r
rownames <- c("math", "science", "history", "English", "law")
colnames <- c("male", "female")
subject_matrix <- matrix(
  c(10,9,11,13,12,18,17,13,6,5),
  nrow = 5, ncol = 2, byrow = TRUE,
  dimnames = list(rownames, colnames)
)
```

- Add row: `rbind()`
- Add column: `cbind()`

ℹ️ Try to add a row to `subject_matrix` for `geography <- c(4,2)` using `rbind` in your Script Editor.

Matrices to Data Frames

- Convert a matrix into a data frame to **mix numeric and non-numeric types**.

```r
matrix_stats <- matrix(
  c(4015,3980,3756,3333,4000,3000,2000,1000),
  nrow = 4, ncol = 2, byrow = TRUE,
  dimnames = list(
    c("freshmen", "sophomores", "juniors", "seniors"),
    c("enrollment", "living on campus")
  )
)

df_stats <- as.data.frame(matrix_stats)
avg.grades <- c("A-","B+","B-","A+")
school_stats <- cbind(df_stats, avg.grades)
```

ℹ️ Try to run the code in your Script Editor.

Data Frames

- Tabular structure with columns of different types
- Similar to spreadsheets

```r
cats <- data.frame(
  coat = c("Persian", "black", "tabby"),
  weight = c(2.1, 5.0, 3.2),
  likes_string = c(TRUE, FALSE, TRUE)
)

animal_sleep <- data.frame(
  ranking = c(4, 1, 2, 3),
  animal = c("koala", "capybara", "camel", "panda"),
  country = c("Australia", "Brazil", "Egypt", "China"),
  avg_sleep_hr = c(20, 3, 6, 12)
)
```

ℹ️ Try to run the code in your Script Editor.

Lists

- Flexible data structure with elements of **different types**
- Can include vectors, data frames, matrices, or other lists

```r
my_list <- list(
  name = "Alice",
  age = 25,
  scores = c(90, 85, 92),
  passed = TRUE
)

my_list$name
my_list[[2]]
```

ℹ️ Try to run the code in your Script Editor.

Arrays

- Extend vectors to two or more dimensions
- All elements must be the same type

```r
arr <- array(1:12, dim = c(2, 3, 2))
arr
arr[1, 2, 1]   # element at row 1, col 2, slice 1
```

Useful for structured multi-dimensional data. However, this is hardly used in most cases.

ℹ️ Try to run the code in your Script Editor.

Importing data

- Create data manually, but usually import from files

Common formats
- **Text**: `.txt` (`readLines()` function)
- **Tabular data**: `.csv`, `.tsv` (`read.table()` function or `read_csv()` function from the `readr` package)
- **Excel**: `.xlsx` (`xlsx` package)
- **Google sheets**: (`googlesheets` package)
- **Statistics program**: SPSS, SAS (`haven` package)
- **Databases**: MySQL (`RMySQL` package)

Example:

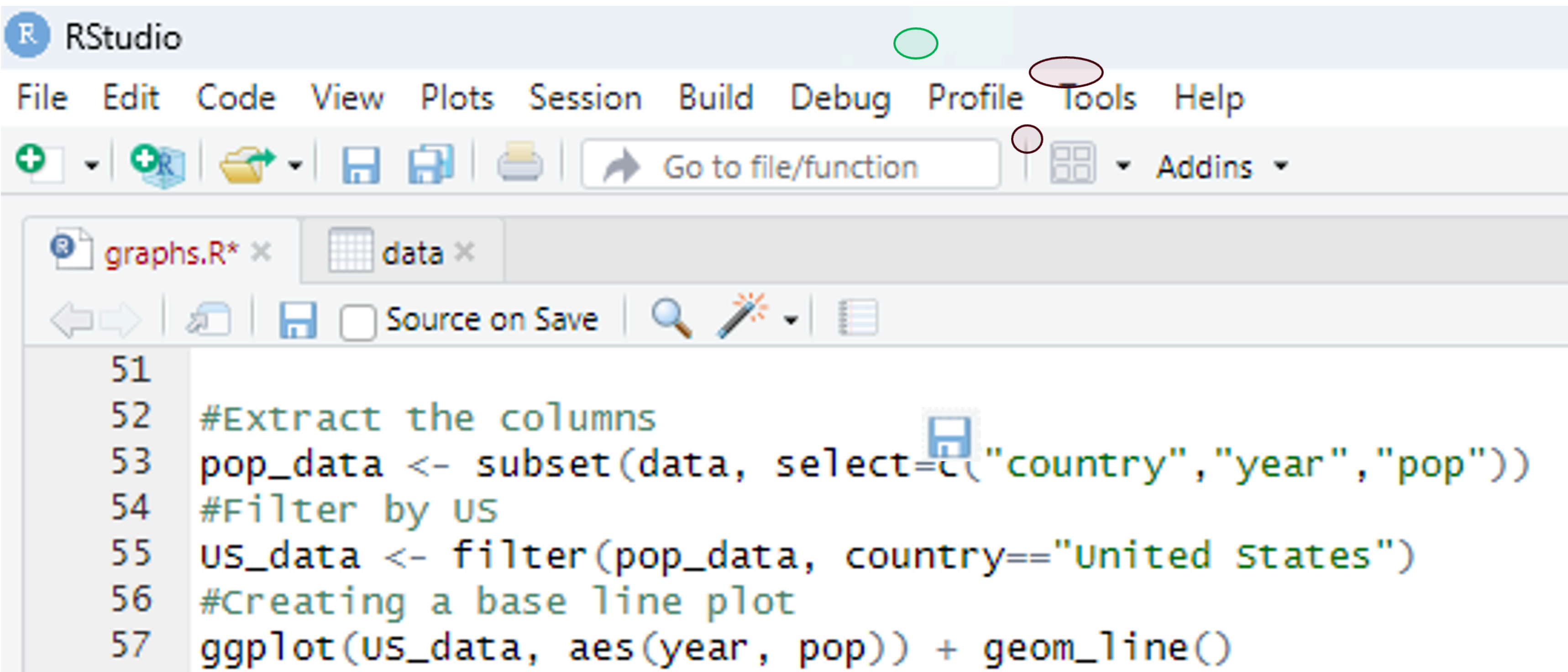
```
csv_data <- read.csv("data/world_data.csv")
```

➡️ Always use forward slashes `/` in file paths.

ℹ️ Try to run the code in your Script Editor.

Saving your script

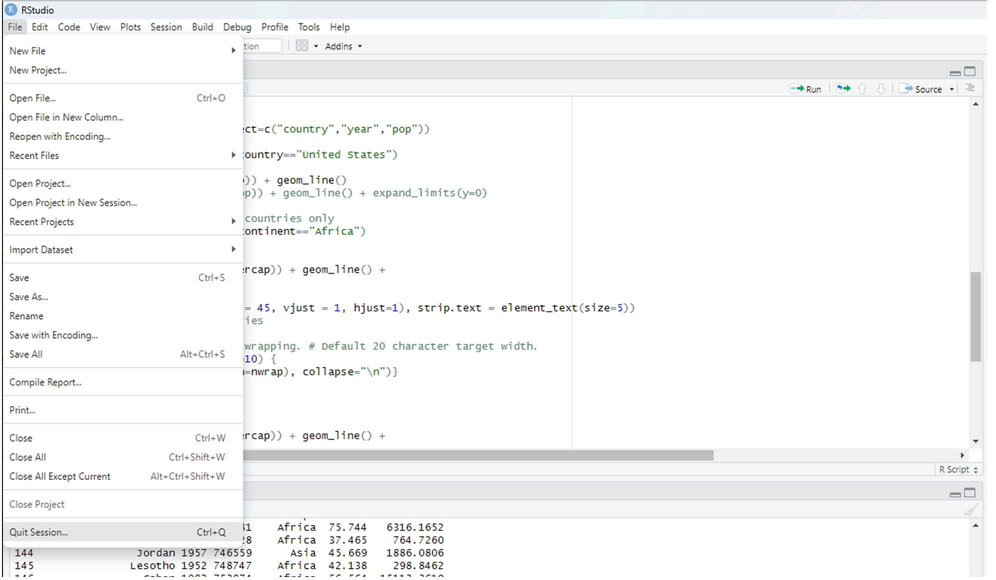
- Save all steps of analysis in `.R` scripts
- Add comments with `#`
- Save with **File → Save** or `Ctrl+S` / `Cmd+S`
- Unsaved scripts show in red with an asterisk `*`



ℹ️ Try to save your script.

Closing an R Session

Options:
- **File → Quit Session**
- `q()` in Console



➡️ Do **not** save the workspace image. Always start fresh for reproducibility.

ℹ️ Try to close and reopen your R Session.

Take home

- R is a **versatile tool for data analysis and visualization** – free, open-source, and supported by a vast community with thousands of packages.
- RStudio **enhances productivity** – organize, edit, and execute code efficiently using the Script Editor, Console, and built-in panes.
- Understand **R objects, data types, and structures** – Vectors, matrices, lists, data frames, and arrays form the building blocks of R programming.
- **Reproducibility** matters – save scripts, comment your code, and avoid saving workspace images to ensure analyses can be repeated.
- Packages **expand functionality** – install, load, and use packages to extend R’s capabilities for importing, manipulating, and visualizing data.

Resources

R Core Team. (2025). R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria. https://www.R-project.org

The official reference for R itself, including installation, base functions, and documentation.

RStudio Team. (2025). RStudio: Integrated Development Environment for R. https://posit.co

Wickham, H., & Grolemund, G. (2017). R for Data Science: Import, Tidy, Transform, Visualize, and Model Data. O'Reilly Media. https://r4ds.had.co.nz

Venables, W. N., Smith, D. M., & the R Core Team. (2023). An Introduction to R. CRAN. https://cran.r-project.org/doc/manuals/r-release/R-intro.pdf

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). An Introduction to Statistical Learning with Applications in R (2nd edition). Springer.

Exercise

Please download the R script with exercises from GitHub and try to complete it.